TECHNICAL USE-CASE GUIDE

NxtGen AI Services

Use cases translated into implementation patterns

A practical engineering guide to the nine AI solution stories in the supplied deck - including data flows, model patterns, platform components, examples, and validation metrics.

From data to measurable outcomes

Use this document to understand where each use case fits, what the technical system must do, and what to test before treating a brochure result as a production baseline.

Prepared from: NxtGen_AI_Services_-_Use_Cases.pdf
Important: use-case results and customer references are claims from the supplied deck and are not independently verified in this guide.

01 | EXECUTIVE VIEW

What the use-case deck demonstrates

The 41-slide deck presents nine AI solution stories across public-sector, automotive, media, telecom, HR, government, and consumer-image workflows. The common theme is not one model; it is a production system that joins data, domain workflows, AI models, controls, and measurable operations.

The nine stories

- Population-scale facial de-duplication and image-quality analysis for an electoral roll.
- AI-assisted procurement and bid evaluation for a government-owned telecom provider.
- Sovereign speech-to-speech assistant for an automobile manufacturer's EV lineup.
- AI document intelligence for contract summarization, comparison, merging, chronology, and Q&A.
- Sales intelligence and revenue forecasting across CRM and media-channel data.
- Field-sales assistant for product, pricing, competitor intelligence, and proposal drafting.
- HR policy assistant connected to an enterprise HRMS through retrieval-augmented generation.
- Conversational revenue forecasting and what-if analysis for a government and enterprise context.
- Sovereign image in-painting for visualizing paint colors in a real home environment.

Use-case lesson: a model is only one component. In every story, the useful outcome also depends on ingestion, data quality, workflow integration, human review, access control, evidence, monitoring, and a repeatable deployment process.

How to read the numbers

Figures such as 90+ crore records, >95% match threshold, 2-3 days reduced to 30-40 minutes, 8 hours reduced to seconds, 500Cr+ revenue, and near-zero errors are claims shown in the supplied deck. They are useful as discovery prompts; request the baseline, measurement method, sample, period, and production evidence before using them as commitments.

02 | SHARED FOUNDATION

The platform pattern behind the case studies

The case studies can be mapped onto a common architecture: governed data, application services, model/runtime infrastructure, workflow integration, and operational controls.

Layer	Responsibilities	Use-case examples
Data and ingestion	Collect structured, semi-structured, image, audio, document, telemetry, CRM, HRMS, and financial data; classify, clean, version, and retain it.	Voter images, tender documents, vehicle telemetry, contracts, Salesforce, policy documents, and revenue datasets.
AI processing	Run computer vision, speech, NLP, retrieval, forecasting, agentic workflows, or image generation against controlled model versions.	Face embeddings, speech-to-speech, RAG, time-series forecasting, document extraction, and in-painting.
Application workflow	Expose results through dashboards, assistants, mobile or vehicle apps, proposal tools, review queues, and decision history.	Bid evaluation dashboard, CarPlay/Android Auto, contract workspace, CXO dashboard, and field-sales assistant.
Controls and evidence	Enforce authorization, privacy, human review, citations, audit trails, versioning, quality gates, and recovery procedures.	Page-level bid evidence, policy-grounded HR answers, electoral quality checks, and contract provenance.
Operations	Deploy, monitor, scale, patch, back up, retrain, and improve the complete service rather than only the model.	High-concurrency image service, population-scale batch processing, and live sales or vehicle experiences.

Common delivery flow

- Define the decision or experience to improve and the human owner of the outcome.
- Map sources, data rights, data quality, update frequency, and retention.
- Select model and runtime based on quality, latency, concurrency, memory, and cost.
- Integrate into the existing workflow, with evidence and human review where decisions are sensitive.
- Use the SAS-F lifecycle to move from bespoke solution engineering and data curation through validation, deployment, and continuous improvement.
- Measure the complete service, then operate it with versioning, alerts, rollback, and continuous improvement.

03 | ARCHITECTURE CHECKLIST

Questions that apply to every use case

Before comparing use cases, establish a common technical language. These questions prevent a successful demo from being mistaken for a production-ready system.

Dimension	Questions to answer
Outcome	What decision, task, or user experience changes? What is the baseline time, quality, cost, and error rate?
Data	Which sources are authoritative? How are permissions, data quality, lineage, retention, and updates handled?
Model	Is the workload classification, extraction, retrieval, generation, speech, forecasting, vision, or a hybrid? Which model versions and licenses are allowed?
Human control	Where must a person validate, approve, correct, or override the system? How is feedback captured?
Performance	What are p95 latency, throughput, concurrency, batch completion, queue time, and scale targets?
Trust	How are evidence, citations, confidence, quality, fairness, privacy leakage, and unsafe outputs checked?
Operations	Who owns the application, data pipeline, model, platform, incident response, recovery, and cost?

A practical benchmark includes the real data path: read -> parse or preprocess -> model -> post-process -> human or system decision -> audit -> response. Synthetic model-only tests are useful, but they are not enough.

01 | USE CASE 1

Population-scale electoral de-duplication and facial quality analysis

The deck describes a national-scale facial recognition pipeline for duplicate voter registration detection, processing 90+ crore records with a >95% match threshold and identifying low-quality, multi-face, and no-face photographs. The story is framed as supporting India's national electoral authority.

Technical view	What the system needs to do	Example implementation
Ingestion	Process a very large image corpus with metadata, partitioning, retries, and quality checks.	Batch jobs read images from partitioned storage and emit status for every record.
Feature extraction	Generate facial embeddings and image-quality signals such as face count, blur, and missing face.	GPU inference workers produce embeddings and quality labels with model/version IDs.
Candidate matching	Compare embeddings at scale and apply a threshold to produce candidate duplicates for review.	Approximate nearest-neighbor index reduces all-to-all comparison cost.
Governance	Protect identity data, limit access, retain evidence, and support human adjudication.	Store match scores, source records, threshold, reviewer outcome, and audit trail.

Implementation bullets

- Treat the match threshold as a candidate-generation parameter, not an automatic legal decision; define review and appeal workflows.
- Measure image-quality errors separately from duplicate-detection quality so poor source images do not contaminate model evaluation.
- Design for restartable batch processing, deduplicated outputs, and checkpointed progress across a population-scale run.
- Use strict access controls, encryption, retention limits, and documented data-protection policies for biometric data.

Measure it with

match precision/recall, false-positive and false-negative rates, review yield, image-quality classification accuracy, records/hour, GPU utilization, retry rate, and end-to-end processing time.

Because this is a high-impact biometric workflow, validate demographic performance, threshold calibration, human oversight, legal basis, and auditability independently of the marketing result.

02 | USE CASE 2

AI-enabled procurement and bid evaluation

The deck describes an AI-powered procurement module for a government-owned telecom provider, reducing vendor assessment from 2-3 days to 30-40 minutes with auditable, evidence-backed scoring across RFP and tender submissions.

Technical view	What the system needs to do	Example implementation
RFP understanding	Extract eligibility, technical, and financial requirements from the RFP or tender.	Document parser creates structured criteria with page and section references.
Bid matching	Compare bidder documents against the extracted criteria and classify each requirement.	Rules plus retrieval and language-model reasoning return Compliant, Non-Compliant, Partially Compliant, or Not Found.
Evidence	Explain every classification with document and page-level references for human validation.	Reviewer opens the cited passage and accepts, edits, or rejects the result.
Decision workflow	Track all bidders, financial gaps, progress, history, reports, and dashboards.	Evaluation state is versioned by RFP, bidder, stage, and reviewer.

Implementation bullets

- Normalize document formats, OCR quality, tables, annexures, and scanned pages before extracting requirements.
- Keep deterministic rules for mandatory eligibility and financial checks; use generative reasoning where interpretation is needed.
- Do not allow an unsupported answer: every score or classification should point to the source text or be marked Not Found.
- Preserve the original submission, extracted representation, model version, prompt, evidence, reviewer decision, and final report.

Measure it with

extraction accuracy, classification precision/recall by requirement type, evidence citation accuracy, reviewer correction rate, processing time per bid, queue time, and audit completeness.

Procurement is a decision-support workflow. Define the human approval point and the legal status of AI-generated classifications before production use.

03 | USE CASE 3

Sovereign speech-to-speech EV companion

The deck describes Tenshi, a speech-to-speech AI companion for an automobile manufacturer's EV lineup, integrated with an internal app, Apple CarPlay, and Android Auto. It lists daily briefings, EV efficiency insights, crash detection, theft alerts, and natural-language trip planning, trained on telemetry data without third-party model APIs.

Technical view	What the system needs to do	Example implementation
Audio loop	Capture speech, detect turn-taking, transcribe, reason, and synthesize a response with low perceived latency.	Streaming speech recognition -> dialogue service -> streaming text-to-speech.
Vehicle context	Use telemetry and vehicle state to ground responses and trigger event-specific experiences.	A typed telemetry contract supplies battery, efficiency, location, and alert context.
Client integration	Deliver a consistent interaction through the internal app, CarPlay, and Android Auto.	Authenticated mobile and in-car clients call a versioned conversation API.
Safety	Separate informational assistance from safety-critical alerts and define fail-safe behavior.	Crash or theft signals use deterministic event paths with clear escalation and human or emergency-service boundaries.

Implementation bullets

- Define latency budgets for wake word, speech recognition, model response, synthesis, and audio playback.
- Minimize and classify telemetry; do not send more vehicle or location data to the model than the task needs.
- Test noisy cabins, multiple languages, accents, interruptions, offline behavior, poor connectivity, and hands-free operation.
- Keep a clear boundary between a conversational suggestion and an authoritative vehicle or emergency action.

Measure it with

time to first audio, end-to-end response latency, word error rate, intent accuracy, interruption recovery, task completion, false-alert rate, offline fallback, and battery/network overhead.

The sovereignty claim is strongest when the deployment, telemetry path, model artifacts, support access, and logs are all documented - not only the model API endpoint.

04 | USE CASE 4

AI-enabled contract management

The deck describes a document-intelligence platform for an Indian passenger vehicle manufacturer's legal and compliance teams. It covers summarization, comparison, merging, chronology tracking, and natural-language Q&A, replacing manual cycles that could take hours with seconds-to-minutes processing.

Technical view	What the system needs to do	Example implementation
Document processing	Ingest contracts, amendments, exhibits, and regulatory documents while preserving structure and provenance.	OCR, layout parsing, clause segmentation, and document/version IDs.
Understanding	Summarize, compare clauses, detect changes, and answer questions grounded in the documents.	Retrieval plus structured extraction returns clause references and confidence.
Chronology	Track events, amendments, obligations, and effective dates across related documents.	Timeline records link an event to a source page and document version.
Secure workspace	Allow authorized legal and compliance users to search, review, merge, and export results.	Role-based access, audit log, retention rules, and review status.

Implementation bullets

- Model contracts as a versioned graph of documents, clauses, parties, events, and obligations rather than a single text blob.
- Keep page-level or paragraph-level citations for every summary, comparison, and answer.
- Test tables, scanned documents, handwritten notes, definitions, cross-references, and conflicting amendments.
- Require legal review for high-risk interpretations; make the platform's output an evidence-backed assistant, not an autonomous legal opinion.

Measure it with

summary factuality, clause-change recall, chronology accuracy, citation coverage, reviewer edit rate, processing time per document, search latency, and access-policy violations.

The reported time savings should be measured against a defined document mix and a quality threshold. Faster text generation is not enough if reviewers must rebuild the chronology manually.

05 | USE CASE 5

Sales intelligence and revenue forecasting

The deck describes the TIMES Impact Platform for an out-of-home media company, integrating Salesforce CRM, TOM data, and 15+ media-channel sources for account insights, seasonality trends, forecasting, and growth recommendations across 500Cr+ revenue, 10,000+ deals, and 50+ sales teams in India.

Technical view	What the system needs to do	Example implementation
Source integration	Unify CRM, TOM, media-channel, bookings, cost, and regional data with common identifiers.	Scheduled ingestion plus change tracking produces a governed sales data model.
Analytics	Generate account-level visibility into revenue, renewals, seasonality, and market opportunities.	Feature pipelines feed dashboards and recommendation services.
Forecasting	Estimate future revenue and expose assumptions, confidence, and scenario changes.	Time-series models are evaluated by segment, horizon, and forecast version.
Leadership UX	Allow sales teams and CXOs to query metrics in natural language.	A conversational dashboard translates approved questions into governed queries.

Implementation bullets

- Create a metric contract for revenue, bookings, pipeline, renewal, cost, and sales-team definitions before model development.
- Reconcile CRM and media-channel identifiers, late-arriving data, currency, time zones, and historical restatements.
- Separate descriptive analytics from causal recommendations; show the data and assumptions behind every recommendation.
- Control natural-language query scope so users cannot bypass row-level or region-level permissions.

Measure it with

forecast error by horizon, bias by segment, data freshness, reconciliation rate, query success, dashboard latency, recommendation acceptance, and revenue impact.

A large revenue footprint increases governance needs. Establish data ownership and forecast accountability before using the assistant in executive decisions.

06 | USE CASE 6

Field-sales assistant

The deck describes an intelligent sales assistant for a global telecom provider's on-ground workforce, providing answers on plans, offerings, and pricing, supporting competitor analysis, and drafting customer-ready Email and WhatsApp proposals.

Technical view	What the system needs to do	Example implementation
Knowledge retrieval	Ground answers in current product, pricing, eligibility, and policy documents.	Versioned retrieval index with effective dates and audience filters.
Field experience	Return concise answers quickly on mobile devices and variable network connections.	Mobile-friendly API with caching, timeout, and offline or degraded-mode behavior.
Market intelligence	Compare approved competitor information without presenting stale or unverified claims as facts.	Source-tagged competitor records with freshness and confidence indicators.
Proposal drafting	Turn approved answers into customer-ready communications for Email and WhatsApp.	Template-controlled generation with review, redaction, and send approval.

Implementation bullets

- Make pricing and offer eligibility data effective-dated so the assistant does not mix old and current plans.
- Use citations or source labels for internal answers and clearly mark competitor intelligence with freshness.
- Separate drafting from sending; require a seller or supervisor to approve customer-facing messages.
- Capture unanswered questions as a feedback queue for content owners and model or retrieval improvement.

Measure it with

answer accuracy, stale-content rate, response latency, seller adoption, correction rate, proposal approval time, conversion rate, and offline success.

For field users, the product is the full interaction loop: fast answer, trusted source, usable draft, and safe send - not only the model's text quality.

07 | USE CASE 7

Intelligent HR agent

The deck describes Ask HR for Godrej & Boyce, an internal assistant for HR policies, travel guidelines, and compliance rules. It integrates with the enterprise HRMS through retrieval-augmented generation and adds an admin intelligence dashboard.

Technical view	What the system needs to do	Example implementation
Policy retrieval	Retrieve the current policy and rule that applies to the employee's question.	HR documents are chunked, tagged by audience and effective date, and indexed.
HRMS integration	Use employee context only when authorized and necessary for the answer.	Identity-aware API retrieves role, location, grade, or leave context under policy.
Grounded response	Answer in natural language with links or citations to policy sources.	The model is constrained to retrieved content and can say it does not know.
Admin intelligence	Surface recurring questions, policy gaps, and workload signals to HR leadership.	Aggregated analytics remove unnecessary employee-level detail.

Implementation bullets

- Apply row and document-level authorization; an employee should not retrieve another employee's HR data.
- Index effective dates, geography, business unit, and policy hierarchy so superseded policies are not retrieved.
- Route exceptions, grievances, payroll changes, and high-impact requests to a human HR process.
- Monitor unanswered and low-confidence questions to improve policies, content, and employee experience.

Measure it with

grounded-answer rate, citation accuracy, policy freshness, escalation rate, employee satisfaction, response latency, access violations, and HR query deflection.

RAG improves grounding but does not by itself provide authorization. Enforce permissions before retrieval and again before returning the answer.

08 | USE CASE 8

Revenue intelligence and conversational forecasting

The deck describes an AI-powered revenue intelligence and forecasting platform for Government of Ajman and BEEAH Group, onboarded through a strategic partnership with Bosch. It uses IBM Granite TTM for time-series forecasting, natural-language queries, what-if analysis, and Excel-to-AI transformation.

Technical view	What the system needs to do	Example implementation
Data model	Convert spreadsheets and multi-source revenue data into a governed analytical model.	Load, validate, map, and version measures such as price, capacity, and employee cost.
Forecasting	Generate projections and scenario analyses with assumptions visible to leadership.	Forecast service stores model version, horizon, features, and scenario parameters.
Conversation	Let leaders ask questions in natural language without creating uncontrolled SQL or spreadsheet formulas.	Intent parser maps approved questions to metric and dimension templates.
Decision support	Compare baseline and what-if scenarios and export a traceable decision view.	Dashboard shows inputs, changes, output ranges, and source data status.

Implementation bullets

- Define a semantic layer for revenue, capacity, headcount, price, and cost before enabling natural-language questions.
- Distinguish forecast uncertainty from scenario assumptions; expose confidence intervals and data freshness.
- Validate spreadsheet transformations against known reports before retiring manual processes.
- Log every scenario query and parameter so leadership decisions remain reproducible.

Measure it with

forecast error, scenario reproducibility, data freshness, query-to-metric accuracy, time to insight, user adoption, and decision impact.

Conversational analytics should be a governed front end to trusted metrics, not an unconstrained replacement for the organization's financial definitions.

09 | USE CASE 9

AI-powered paint visualisation

The deck describes a sovereign in-painting solution that lets customers visualize paint colors in their own home photos, with claims of reflection preservation, high-concurrency performance, lower TCO, near-zero errors, and no dependency on third-party AI APIs. It also shows a comparison against Google Nano Banana and GPT Image 1.

Technical view	What the system needs to do	Example implementation
Image preparation	Accept a customer photo, identify the target surfaces, and preserve non-target regions.	Segmentation or masking creates a reproducible edit mask and quality checks the input.
In-painting	Change the selected surface while preserving lighting, geometry, texture, and reflections.	Conditioned image pipeline generates the edited region and composites it with the source.
Quality and safety	Reject poor images, incorrect masks, or outputs that alter protected regions.	Automated visual checks plus human or customer confirmation for edge cases.
Scale	Serve many interactive image requests with predictable latency and cost.	GPU inference workers use queueing, caching, concurrency controls, and observability.

Implementation bullets

- Separate segmentation accuracy from color-rendering quality; a wrong mask can look like a model failure.
- Test windows, tube lights, mirrors, water containers, furniture, shadows, and mixed surfaces because reflections are part of realism.
- Define unacceptable edits such as changing furniture, people, electrical fixtures, or structural boundaries.
- Compare against external APIs using the same inputs, masks, quality rubric, latency target, and cost assumptions.

Measure it with

mask IoU, color fidelity, reflection preservation, protected-region error, user acceptance, p95 image latency, concurrency, failure rate, and cost/image.

Claims such as near-zero errors and outperforming named external systems need a repeatable benchmark set, scoring rubric, and independent review before they become procurement criteria.

13 | CROSS-CUTTING DESIGN

One operating model for very different AI workloads

The use cases vary widely, but the same production controls keep recurring. Standardize the controls and specialize the model and workflow.

Control	Why it matters	Example evidence
Data provenance	AI output quality depends on the source data, its freshness, and the transformation path.	Lineage, source IDs, effective dates, dataset version, data-quality report.
Model and prompt versioning	A result cannot be reproduced or rolled back if the model, prompt, or retrieval settings are unknown.	Model registry, prompt hash, runtime version, index version, release record.
Evidence and explainability	Decision-support workflows need a reviewer to see why the system produced a result.	Page citation, source image, match score, clause reference, forecast assumptions.
Human-in-the-loop	Sensitive or high-impact outcomes need an owner who can approve, correct, or override.	Review queue, approval state, correction reason, escalation path.
Observability	Production quality can drift even when infrastructure is healthy.	Quality metrics, latency, utilization, data freshness, drift, error rate, alerts.
Recovery	The service includes data, indexes, configuration, models, and workflow state - not just containers.	Backup, restore test, RPO/RTO, rollback, incident runbook.

A reusable reference pattern

- Private or sovereign data zone -> curated dataset or index -> model/runtime service -> governed API or workflow -> evidence and audit -> monitoring and improvement.
- Use the same identity, deployment, logging, backup, and incident controls across workloads even when the models differ.
- Keep workload-specific acceptance metrics: match quality for biometrics, citation accuracy for documents, WER for speech, forecast error for revenue, and visual fidelity for images.

14 | POC AND PROCUREMENT CHECKLIST

How to turn a case study into a decision

Use this sequence when a use case resembles one of the stories in the deck. It creates a clear path from a compelling demo to a measurable, supportable production service.

Step	Output
1. Define the baseline	Current time, quality, error, cost, volume, and human effort for the existing process.
2. Define the AI decision	What the system predicts, extracts, generates, recommends, or automates - and what it must never do.
3. Prepare representative data	Realistic data mix, edge cases, permissions, data-quality issues, and an evaluation set kept separate from development.
4. Benchmark end to end	Quality, latency, throughput, concurrency, failure behavior, evidence coverage, and cost for the complete workflow.
5. Test governance	Access, privacy, licensing, fairness, human review, auditability, retention, and recovery evidence.
6. Contract operations	Service levels, ownership, support boundaries, updates, rollback, model refresh, and exit or portability plan.

Bottom line

The use-case deck shows breadth: biometrics, documents, speech, forecasting, sales, HR, and image generation. The reusable technical lesson is to build a governed service around the model. The right next step for any one of these patterns is a representative proof of concept with explicit data, quality, performance, security, ownership, and cost criteria.

Source boundary

All customer names, sector descriptions, workload volumes, time savings, model names, thresholds, performance statements, and outcome claims in this guide are derived from the supplied use-case deck. Confirm them against current customer-approved references, benchmark artifacts, security documents, service levels, and commercial scope before relying on them for procurement or public claims.

Contact shown in source deck: marketing@nxtgen.com